The Economic Value of Knowledge Geometry: Topological Opportunities and Breakthrough Innovation in Scientific Networks

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Abstract

This paper studies whether the geometry, topology, and local spectra of scientific knowledge networks shape breakthrough innovation. Using the OGBN-MAG subset of the Microsoft Academic Graph, I construct a dynamic scientific knowledge complex from paper-topic assignments, citation links, authorship, venues, and publication years. The central topological object is boundary-completion potential: whether a paper's topic set spans triangular boundaries already present in the prior topic graph. I also compute local clique-complex Betti numbers and Laplacian spectral metrics on each paper's prior topic-induced graph. Papers published between 2011 and 2016 are linked to their three-year forward citations and classified as breakthrough papers when they fall in the top five percent of their publication-year citation distribution. The main evidence shows that boundary-completion potential and local spectral connectivity are associated with future scientific impact after controlling for authorship, topic breadth, references, year fixed effects, and venue-group fixed effects. A causal-style exposure design uses lagged, leave-venue-out topology shocks from papers in the same fields of study: external frontier opening strongly shifts a focal paper's own topological opportunity and predicts higher future citations and breakthrough probability. The results suggest that breakthrough science is not only a function of research scale or centrality, but also of the high-order structure and diffusion geometry of the knowledge space in which researchers search.

Keywords: science of science, innovation, topology, spectral graph theory, Microsoft Academic Graph, OGBN-MAG

1. Introduction

Where do breakthrough innovations come from? A large literature in economics, management, and the science of science emphasizes cumulative knowledge, recombination, collaboration networks, and the uneven distribution of scientific opportunity (Uzzi et al., 2013; Fortunato et al., 2018; Acemoglu et al., 2016). Yet most empirical measures of scientific opportunity remain low-order: publication counts, citation counts, team size, field fixed effects, or pairwise network centrality. These measures are informative, but they do not directly characterize the high-order geometry of the knowledge space in which research ideas are combined.

This paper develops and tests a simple claim: scientific breakthroughs are more likely when researchers occupy topological opportunities in the prior knowledge space. A topological opportunity is a location where existing topics are near enough to be meaningfully connected, but the higher-order knowledge combination has not yet been stabilized. In such locations, a paper can add more than another node to the scientific graph. It can close a gap, fill a boundary, or create a bridge between previously separated knowledge regions.

The empirical setting is the OGBN-MAG subset of the Microsoft Academic Graph (Sinha et al., 2015; Hu et al., 2020). The data contain 736,389 papers, 1,134,649 authors, 8,740 institutions, 59,965 fields of study, 5.4 million citation edges, and 7.5 million paper-topic edges. For each paper, I observe publication year, venue label, authorship, references, and field-of-study assignments. I build a dynamic knowledge complex year by year. Before each paper is published, I compute whether its topics connect previously unobserved pairs, whether they span triangular boundaries in the prior topic graph, and whether they lie on negatively curved topic-pair bridges.

The core analysis uses papers published from 2011 through 2016, which provides a three-year forward citation window for all papers in the analytic sample of 485,819 papers. The outcome is either log three-year forward citations or an indicator for being in the top five percent of the publication-year citation distribution. The main specifications include controls for authors, topic breadth, references, year fixed effects, and venue-group fixed effects. To move beyond predictive association, I also construct a lagged external topology exposure that measures whether the focal paper's fields experienced topological frontier opening in the previous year outside the focal venue group.

The results are preliminary but encouraging in a specific sense. Regression estimates show that the boundary-completion index is positively associated with future scientific impact. Component specifications indicate that novel topic-pair formation is more closely related to tail breakthroughs than to average citations, while local Laplacian connectivity and spectral entropy are positively associated with both outcomes. Local β_1 holes are rare in this paper-level construction and do not carry independent explanatory power. The external-exposure design shows the same pattern from a more causal angle: when adjacent fields experience leave-venue-out frontier opening in the previous year, focal papers are more likely to occupy topological opportunities and later receive more citations. The value of topology and spectral graph theory is therefore not a small black-box forecasting gain. It is that they identify high-order and diffusion-relevant structures of scientific opportunity that are invisible in scale-based measures.

The paper contributes to three literatures. First, it adds to economics of innovation by proposing an ex ante measure of opportunity in scientific search. Second, it contributes to science-of-science research on recombination and breakthrough discovery by moving from pairwise novelty to high-order topology. Third, it connects topological data analysis and spectral graph theory (Edelsbrunner et al., 2002; Zomorodian and Carlsson, 2005; Chung, 1997; Forman, 2003) to an empirical setting where the objects have economic meaning: scientific papers, authors, fields, and future impact.

2. Related Works

2.1. Innovation, recombination, and the direction of search

This paper is related first to the economics of innovation and knowledge recombination. A central idea in this literature is that new ideas are produced by combining existing ideas, but that not all combinations have the same value. Weitzman (1998) formalizes growth as a recombinant process, while Fleming (2001) shows that new technological combinations increase uncertainty and upper-tail outcomes. Work on local search and boundary-spanning exploration emphasizes that search is shaped by technological and organizational boundaries (Rosenkopf and Nerkar, 2001). Science can also guide technological search by helping inventors navigate complex combinatorial spaces (Fleming and Sorenson, 2004). In science itself, atypical combinations are associated with high impact (Uzzi et al., 2013), while research strategies trade off tradition and risky innovation (Foster et al., 2015). Related work also studies the burden of expanding knowledge and the network structure of innovation across technology classes (Jones, 2009; Acemoglu et al., 2016). This literature motivates the view that scientific search is structured by the existing knowledge environment.

This paper differs from prior recombination work in the object of measurement. Existing measures often focus on pairwise novelty, atypical combinations, field positions, or network centrality. I instead model the prior knowledge space as a dynamic high-order object. The novelty is to measure whether a paper fills a boundary, occupies a local topological gap, or lies in a spectrally connected frontier region. The contribution is therefore not only to say that unusual combinations matter, but to distinguish between isolated novelty and frontier completion.

2.2. Science of science and bibliographic networks

The paper also contributes to the science-of-science literature. Fortunato et al. (2018) synthesize evidence that scientific production is shaped by teams, institutions, careers, fields, and networks. A large body of work uses collaboration and citation networks to study the organization of scientific production, including the structure of collaboration networks (Newman, 2001), the growing role of teams (Wuchty et al., 2007), paper-level citation dynamics (Wang et al., 2013), and scientist-level impact trajectories (Sinatra et al., 2016). Large-scale bibliographic graphs such as Microsoft Academic Graph and OGBN-MAG make it possible to measure these structures at scale (Sinha et al., 2015; Hu et al., 2020). Existing work often represents science as citation, collaboration, or topic networks, and uses these networks to study scientific influence, diffusion, collaboration, and field structure.

This paper uses the same graph-based view of science, but differs in how the graph is converted into an empirical object. Rather than treating MAG only as a citation graph or a paper-author-field network, I construct a dynamic topic complex in which each paper can add edges, simplices, and local spectral structure to the prior knowledge space. This makes it possible to ask a different question: not only whether a paper is central or novel, but whether it changes the high-order geometry of the scientific frontier.

2.3. Topological, spectral, and geometric methods

Finally, the paper connects innovation measurement to topological data analysis, graph geometry, and spectral graph theory. Persistent homology provides tools for measuring connected components, cycles, and higher-order cavities in filtered spaces (Edelsbrunner et al., 2002; Zomorodian and Carlsson, 2005). Broader introductions to topological data analysis emphasize that topology can summarize the shape of complex data beyond pairwise distances or centrality measures (Ghrist, 2008; Carlsson, 2009). Spectral graph theory studies how Laplacian eigenvalues summarize connectivity and diffusion in graphs (Chung, 1997). Discrete curvature offers another way to identify bridge-like or high-load regions of a graph (Forman, 2003). These tools are widely used to describe the shape and connectivity of complex data.

The distinction in this paper is that these mathematical tools are given an economic interpretation and linked to scientific outcomes. Betti numbers and persistence diagrams are used to describe knowledge holes, Laplacian spectra are used to measure local diffusion capacity, and Forman curvature is used to identify bridge-like topic pairs. The empirical contribution is to show that these measurements are not only descriptive summaries of a graph. They define innovation regimes, predict future scientific impact, and support a causal-style exposure design based on lagged external topology shocks.

3. Mathematical Toolkit and Conceptual Framework

This section introduces the mathematical objects used in the paper. The goal is not to import topology as a black-box forecasting device, but to translate several tools from applied mathematics into economic measurements of scientific opportunity. Each object has a direct empirical interpretation: graphs describe pairwise knowledge connections, simplicial complexes describe higher-order combinations, Betti numbers describe holes or disconnected pieces in the knowledge space, Laplacian spectra describe local diffusion capacity, and curvature identifies bridge-like topic pairs.

3.1. Graphs and simplicial complexes

A graph records pairwise relations. In this paper, a topic graph connects two fields of study if they have appeared together in prior papers. This is useful, but it loses information about higher-order combinations. A paper with three fields does not merely create three pairwise links; it may also create a three-way research object. A simplicial complex is the mathematical structure that keeps track of such higher-order combinations. Nodes are fields, edges are two-field combinations, triangles are three-field combinations, and higher-dimensional simplices represent larger topic bundles. This is the reason topology is useful here: it lets the empirical object move from pairwise novelty to high-order recombination.

3.2. Homology, persistence, spectra, and curvature

Homology summarizes the shape of a complex. The Betti number β_0 counts connected components. β_1 counts loops or holes. β_2 counts higher-dimensional cavities. In economic terms, these quantities describe whether a local knowledge neighborhood is fragmented, whether there are missing higher-order combinations, and whether the neighborhood contains unresolved structure. Persistent homology extends this idea across a filtration, asking which holes remain visible as weaker knowledge ties are gradually admitted.

Spectral graph theory provides a complementary language for diffusion. The graph Laplacian summarizes how connected a local topic neighborhood is. Its second-smallest eigenvalue, $\lambda_2(L)$, is the algebraic connectivity of the graph: higher values indicate that the neighborhood is less fragmented and that information can move through more routes. Spectral entropy measures how dispersed the local diffusion modes are. Discrete curvature adds a bridge interpretation. In the Forman curvature proxy used here, very negative curvature marks topic pairs that connect high-degree regions and therefore resemble bottlenecks or bridges in the knowledge graph.

The economic mapping is therefore straightforward. Topological opportunity measures whether a paper occupies an unresolved high-order gap. Spectral connectivity measures whether nearby knowledge is organized enough for the new combination to diffuse. Curvature measures whether a paper uses bridge-like parts of the knowledge space. The empirical claim is not that these mathematical objects are intrinsically valuable; it is that they provide ex ante measurements of scientific search environments that standard scale, venue, and topic-count controls do not capture.

3.3. Paper-level measures

Let $G_t = (V_t, E_t)$ be the cumulative topic co-occurrence graph before year t. A node is a field of study and an edge indicates that two fields have appeared together in at least one prior paper. A paper i published in year

t has a topic set $S_i \subset V_t$. The paper can be interpreted as adding local structure to the prior knowledge graph.

The first component of opportunity is pairwise novelty:

$$N_i = \frac{1}{\binom{|S_i|}{2}} \sum_{\{u,v\} \subset S_i} \mathbf{1}\{(u, v) \notin E_t\}. \tag{1}$$

This measure captures whether the paper combines topics that had not previously appeared together.

The second component is boundary-completion potential. For triples of topics, define:

$$B_i = \frac{1}{\binom{|S_i|}{3}} \sum_{\{u,v,w\} \subset S_i} \mathbf{1}\{(u, v), (u, w), (v, w) \in E_t\}. \tag{2}$$

When the three boundary edges already exist, the new paper supplies a higher-order simplex over a previously established triangular boundary. This is a local, computable analogue of filling a one-dimensional hole in a knowledge complex.

I also compute local Betti numbers on the clique complex induced by S_i in the prior graph. Let $C_k(S_i)$ be the vector space over $\mathbb{F}_2$ generated by k-simplices in the local clique complex, and let $\partial_k : C_k \to C_{k-1}$ be the boundary operator. The local Betti numbers are:

$$\beta_k(S_i) = \dim \ker(\partial_k) - \dim \operatorname{im}(\partial_{k+1}). \tag{3}$$

In the implementation, $k = 0, 1, 2$ are computed exactly on each paper's local topic complex. This gives a small-scale TDA object rather than a verbal topology metaphor.

The spectral component uses the graph Laplacian of the same prior topic-induced graph. Let A_i be the local adjacency matrix, D_i its degree matrix, and:

$$L_i = D_i - A_i. \tag{4}$$

I compute the second-smallest eigenvalue $\lambda_2(L_i)$, the adjacency spectral radius, and Laplacian spectral entropy. $\lambda_2(L_i)$ measures local algebraic connectivity; spectral entropy captures how dispersed the local diffusion modes are.

The final graph-geometric bridge measure is based on Forman curvature. For an unweighted edge (u, v) in the prior topic graph, a simple Forman-Ricci curvature proxy is:

$$F(u, v) = 4 - d(u) - d(v), \tag{5}$$

where $d(\cdot)$ is prior topic degree. Highly negative values identify edges that connect high-degree regions and therefore act as bridge-like, high-load parts of the knowledge geometry. For paper i, I average negative curvature over its already existing topic pairs.

The main topological opportunity index is standardized boundary closure:

$$TO_i = z(B_i). \tag{6}$$

Pairwise novelty, local Betti numbers, Laplacian spectral metrics, and negative Forman curvature are retained as decomposition variables. The object is intentionally ex ante: all components are computed using only knowledge observed before the paper's publication year.

4. Data

The analysis uses OGBN-MAG, a heterogeneous academic network extracted from Microsoft Academic Graph and distributed by the Open Graph Benchmark. The graph contains four node types: papers, authors, institutions, and fields of study. It also contains four directed relation types: authors are affiliated with institutions, authors write papers, papers cite papers, and papers have fields of study.

Table 1: Dataset and analytic sample

Quantity	Count
Papers	736,389
Authors	1,134,649
Institutions	8,740
Fields of study	59,965
Citation edges	5,416,271
Paper-topic edges	7,505,078
Author-paper edges	7,145,660
Analytic papers	485,819

The paper-level outcome is three-year forward citations. For a paper i published in year t, I count citations from papers published in years $t+1$, $t+2$, and $t+3$. The breakthrough indicator equals one if the paper is in the top five percent of the forward-citation distribution among papers published in the same year.

5. Innovation Regimes in the Knowledge Complex

The central mechanism is not that topology is a generic predictor of citations. It is that topology and spectral connectivity jointly define different regimes of scientific search. A paper can be novel in a disconnected part of the knowledge space, or it can be embedded in a well-connected but already routine core. These locations have different economic meanings even if they have similar topic counts or venue labels.

I classify papers along two ex ante dimensions. The horizontal dimension is topological opportunity, measured by standardized boundary-completion potential. It captures whether a paper occupies a location where prior topic pairs have already formed a boundary that can be filled by a higher-order

Table 2: Summary statistics

Variable	Mean	Std. dev.	P25	Median	P75
Three-year forward citations	4.885	11.639	1.000	2.000	5.000
Breakthrough indicator	0.051	0.221	0.000	0.000	0.000
Novel topic-pair share	0.127	0.152	0.000	0.067	0.200
Boundary-completion potential	0.725	0.273	0.500	0.800	1.000
Negative Forman curvature	7444.968	4242.967	3693.286	6954.364	10835.667
Topological opportunity index	0.000	1.000	-0.827	0.274	1.008
Local β_0	1.056	0.276	1.000	1.000	1.000
Local β_1	0.001	0.033	0.000	0.000	0.000
Local β_2	0.000	0.002	0.000	0.000	0.000
Local $\lambda_2(L)$	3.865	1.913	2.000	4.000	6.000
Local spectral radius	4.453	0.643	4.162	4.702	5.000
Local spectral entropy	1.556	0.121	1.554	1.598	1.609
Authors	8.895	91.279	2.000	4.000	6.000
Topics	10.319	1.397	10.000	11.000	11.000
References	6.212	8.791	1.000	3.000	8.000

combination. The vertical dimension is spectral connectivity, measured by the average of standardized local $\lambda_2(L)$, spectral radius, and spectral entropy. It captures whether the local knowledge neighborhood has enough connectivity and diffusion capacity for a new combination to be absorbed.

This produces four innovation regimes. Low topology and sufficient spectral connectivity corresponds to routine core science: the neighborhood is connected, but the focal paper does not fill a major unresolved boundary. High topology and low spectral connectivity corresponds to isolated novelty: the paper enters a structurally unusual location, but the neighborhood is too disconnected for easy diffusion. Low topology and low spectral connectivity corresponds to fragmented search. The key regime is high topology and sufficient spectral connectivity, which I call frontier completion. These are papers that enter unresolved parts of the knowledge complex while remaining close enough to connected knowledge neighborhoods for diffusion and evaluation.

Figure 1 visualizes this regime structure. The left panel splits papers by high and low topological opportunity and by low versus sufficient spectral connectivity, then reports the share that become top-five-percent break-

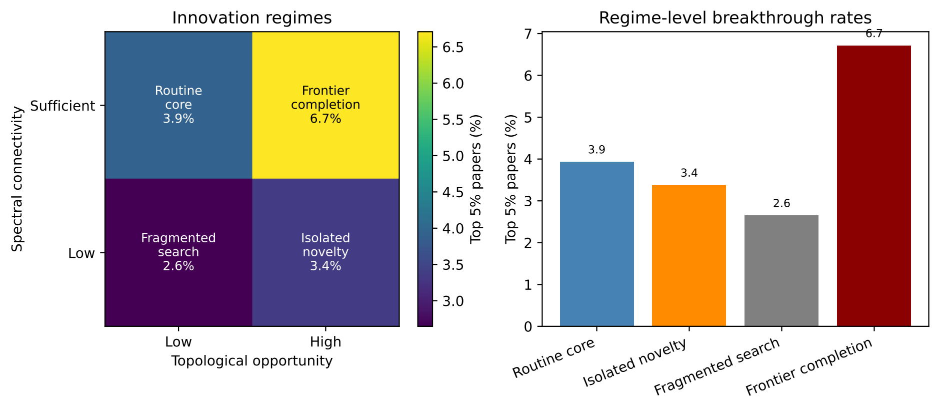

Figure 1: Innovation regimes in the knowledge complex

through papers in each cell. The right panel reports the same four regime-level breakthrough rates as bars. The intended interpretation is not that all novelty is valuable. Breakthrough science is concentrated where unresolved high-order structure is paired with enough spectral connectivity for the idea to travel.

6. Empirical Strategy

The baseline regression is:

$$Y_{i,t+3} = \alpha + \beta TO_{i,t} + X'_{i,t}\gamma + \delta_t + \mu_{v(i)} + \varepsilon_i, \tag{7}$$

where $Y_{i,t+3}$ is either log three-year forward citations or the breakthrough indicator. $TO_{i,t}$ is the topological opportunity index. $X_{i,t}$ includes log authors, log topic count, and log references. δ_t denotes publication-year fixed effects. $\mu_{v(i)}$ denotes venue-group fixed effects for the largest venue groups and an omitted group for all remaining venues.

The main causal-style design exploits external frontier opening in adjacent knowledge neighborhoods. For each paper, define:

$$Z_{i,t-1}^{-v} = \frac{1}{|S_i|} \sum_{f \in S_i} \overline{TO}_{f,t-1}^{-v(i)}, \tag{8}$$

where $\overline{TO}_{f,t-1}^{-v(i)}$ is the mean topological opportunity of papers published in field f in the previous year, excluding papers from the focal paper's venue group. This variable is lagged and leave-venue-out: it captures whether the surrounding scientific frontier opened before the focal paper was published and outside the focal publication venue.

I estimate two specifications:

$$TO_{i,t} = \alpha + \pi Z_{i,t-1}^{-v} + X_{i,t}'\gamma + \delta_t + \mu_{v(i)} + u_i, \tag{9}$$

$$Y_{i,t+3} = \alpha + \rho Z_{i,t-1}^{-v} + X_{i,t}'\gamma + \delta_t + \mu_{v(i)} + \varepsilon_i. \tag{10}$$

The first equation asks whether external frontier opening shifts the focal paper's own topological opportunity. The second is a reduced-form outcome equation. The identifying assumption is not full random assignment. It is that, conditional on publication year, venue group, and paper-level controls, topology changes produced by other venues in the same fields proxy for external knowledge-supply shocks rather than focal-paper quality alone.

7. Results

Figure 2 shows the evolution of the knowledge complex. Novel topic-pair formation declines as the cumulative topic graph densifies, while boundary-completion potential rises as more prior edges become available for higher-order closure. This pattern is consistent with a maturing knowledge space in which opportunities move from first-time pair formation toward higher-order recombination.

Figure 3 provides a dedicated TDA visualization for an active frontier band of the topic complex: fields ranked 101–180 by frequency. The most fre-

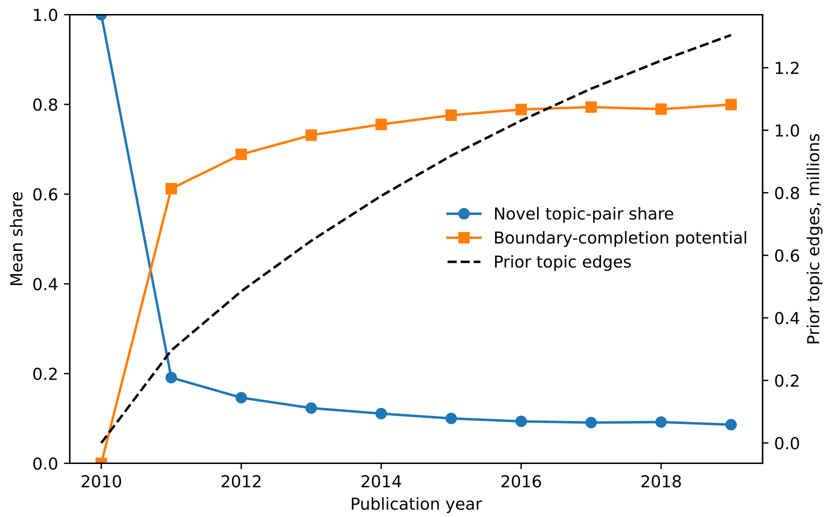

Figure 2: Evolution of topic-pair novelty and boundary-completion potential

quent core fields are already saturated, so this second tier is more informative about unresolved structure. The filtration runs from strong to weak knowledge ties: the highest-frequency topic-pair edges enter first, weaker bridges enter later, and triangles enter when genuine three-topic co-occurrences provide higher-order closure. The Betti curves summarize how components merge and one-dimensional holes appear as weaker bridges are admitted, while the persistence diagram and barcode show the lifetimes of the longest-lived knowledge holes.

Figure 4 shows the paper-level local homology and spectral graph measures. Local β_1 is sparse, which is informative: in this paper-level construction, the prior topic neighborhoods are usually connected and triangle-rich. By contrast, local algebraic connectivity and spectral entropy vary systematically over time, giving usable variation for spectral graph analysis.

Figure 5 bins papers by the topological opportunity index. Papers in higher-opportunity bins receive more future citations and are more likely to become breakthrough papers. The relationship is not a proof of causality, but it is the first empirical fact the paper seeks to explain.

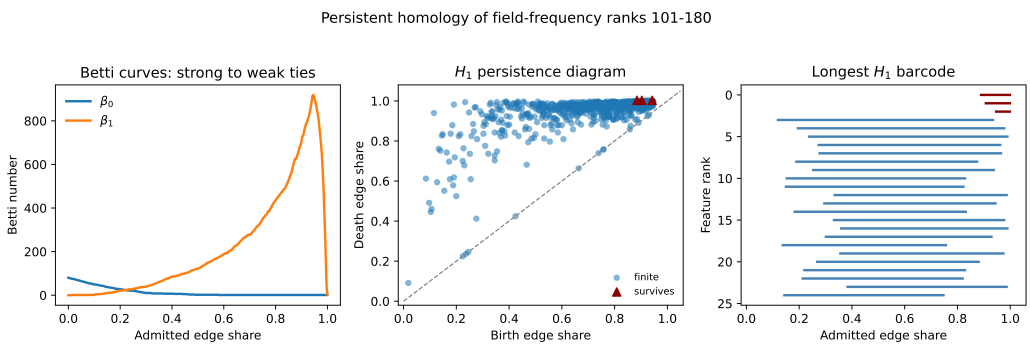

Figure 3: Persistent homology of the scientific knowledge complex

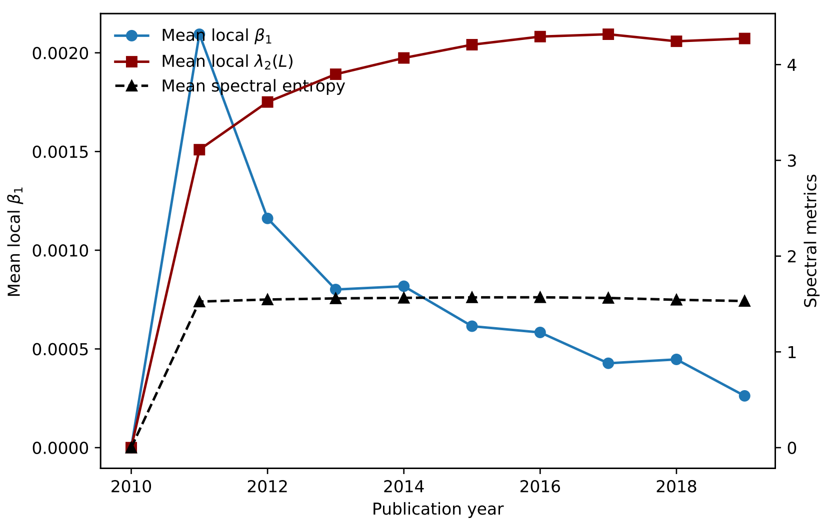

Figure 4: Local TDA and spectral graph measures

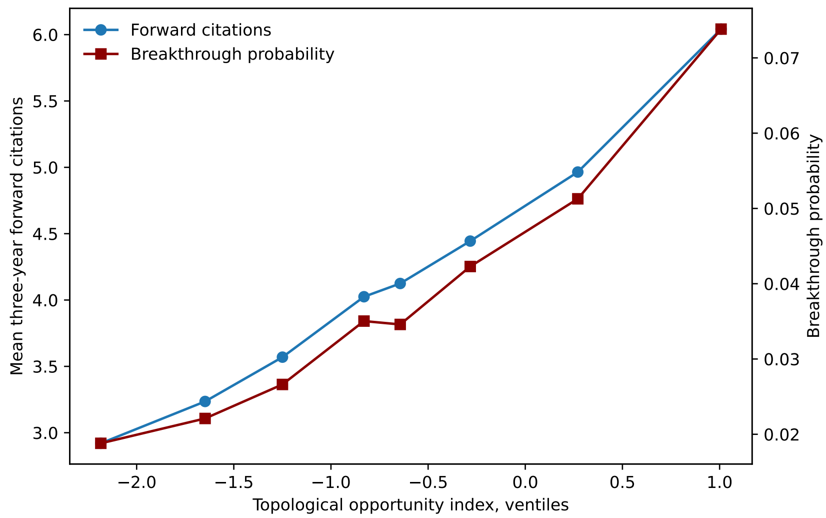

Figure 5: Topological opportunity and future scientific impact

Table 3: Topology, geometry, and future scientific impact

	(1)	(2)	(3)	(4)	(5)	(6)
TO index		0.029***			0.002***	
		(0.001)			(0.000)	
Novel pairs			-0.014***			0.004***
			(0.003)			(0.001)
β_1 holes			-0.000			-0.000
			(0.001)			(0.000)
$\lambda_2(L)$			0.008***			0.006***
			(0.003)			(0.001)
Spectral entropy			0.012***			0.001**
			(0.002)			(0.000)
Negative curvature			-0.013***			-0.001***
			(0.001)			(0.000)
Outcome	Log cites	Log cites	Log cites	Top 5%	Top 5%	Top 5%
Specification	Base	+ TO	+ TDA/Spec.	Base	+ TO	+ TDA/Spec.
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Venue-group FE	Yes	Yes	Yes	Yes	Yes	Yes
N	485,819	485,819	485,819	485,819	485,819	485,819
R^2	0.309	0.310	0.310	0.143	0.143	0.143

Notes: HC1 robust standard errors in parentheses. The analytic sample contains papers published from 2011 through 2016, allowing a three-year forward citation window. Topology variables are standardized within the analytic sample. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 3 reports the main regression evidence. The boundary-completion index is positively associated with both log forward citations and the breakthrough indicator. Component specifications show a useful distinction: first-time topic-pair novelty is negatively associated with average citations but positively associated with the probability of reaching the top tail. Local $\lambda_2(L)$ and spectral entropy are positive in both outcome models, consistent with the interpretation that well-connected local knowledge neighborhoods diffuse more effectively. Local β_1 holes are too sparse to explain much variation in this draft.

Table 4: External topology exposure and scientific impact

Design	Outcome	Coefficient	SE	N
First stage	Topological opportunity	1.912***	(0.004)	485,818
Reduced form	Log cites	0.077***	(0.004)	485,818
Reduced form	Top 5%	0.012***	(0.001)	485,818

Notes: The treatment variable is a standardized external topology exposure: for each focal paper, the mean prior-year topological opportunity of papers that share its fields of study, excluding papers from the same venue group. All specifications include log authors, log topics, log references, publication-year fixed effects, and venue-group fixed effects. The first-stage outcome is the focal paper's own topological opportunity. Reduced-form outcomes are log three-year forward citations and the top-five-percent breakthrough indicator. HC1 robust standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 4 reports the external-exposure evidence. A one-standard-deviation increase in lagged leave-venue-out topology exposure strongly predicts the focal paper's own topological opportunity. The same exposure is positively associated with both log three-year forward citations and the top-five-percent breakthrough indicator. This does not make the paper a fully causal natural experiment, but it changes the empirical object from "can topology forecast impact?" to "do external changes in nearby knowledge geometry shift later scientific outcomes?"

8. Discussion

The main interpretation is that scientific knowledge has economically meaningful shape. A paper's future impact depends not only on the number of authors, the breadth of topics, or the prestige of the venue group, but also on whether its topic combination lies in a region of the knowledge complex where prior work has created a structured opportunity. Pairwise novelty captures unexplored combinations. Boundary-completion potential captures higher-order closure. Local Betti numbers capture small-scale homology. Laplacian eigenvalues capture diffusion geometry. Negative curvature captures bridge-like locations in the topic graph.

This interpretation connects naturally to models of recombinant innovation. If researchers search locally in knowledge space, then the distribution of opportunities is shaped by prior discoveries. Some areas are saturated; others contain gaps that are visible only when the knowledge space is represented as a high-order object rather than as a list of fields or citations. Topology provides a vocabulary for those gaps.

The current draft has three important limitations. First, the outcome is scientific impact rather than direct economic value. Citations are a meaningful measure of scientific influence, but a top-journal economics version should link papers to patent citations, firm innovation, or regional outcomes. Second, the current TDA measures are exact only on local paper-level clique complexes; they are not yet persistent homology over a global filtration of the MAG graph. Third, the external-exposure design improves causal interpretation but does not by itself deliver random assignment. Future versions should strengthen the design by exploiting differential exposure across institutions, regions, funding shocks, or patent-linked technology classes.

9. Conclusion

This paper proposes a new way to measure scientific opportunity. Instead of asking only how many papers, authors, citations, or topics are present, it asks where a paper lies in the geometry and topology of the prior knowledge space. Using OGBN-MAG, the preliminary evidence shows that ex ante topological opportunities predict future breakthrough science.

The broader implication is that innovation policy and science strategy may benefit from measuring the shape of knowledge. If high-value discoveries often arise from topological gaps, then institutions and funding agencies should care not only about field size or recent citation momentum, but also about the latent geometry of recombination.

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